Aileron Control System – General Data

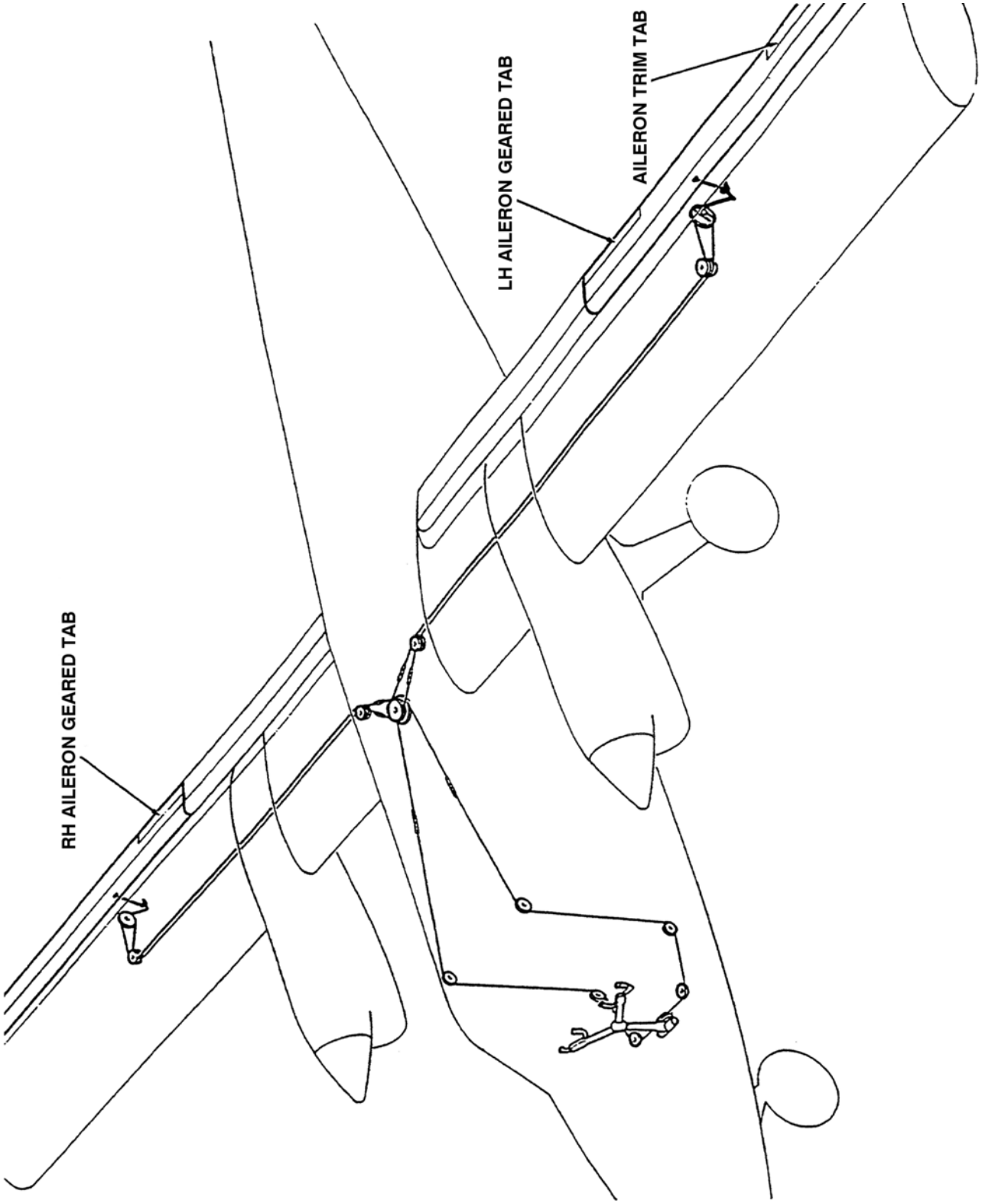
1. General

Rotary movement of the control column aileron control wheels, is transmitted by a chain and sprocket mechanism to a quadrant at the base of the column. Cables extending from the control column quadrant, are routed under the flight compartment floor and up both sides of the forward face of the flight compartment/cabin bulkhead, to transmit motion to an aileron quadrant located in the cabin roof aft of the wing front spar. From the aileron quadrant, a separate cable circuit is routed through each wing to linkage quadrants, located at the left and right second outboard fore flap hinge arm attachment brackets. A push-pull rod connects between each linkage quadrant and a bellcrank lever on the respective fore flap hinge arm. The ailerons attach to the trailing edge of the corresponding outboard flap by four aileron arms protruding from the outboard fore flap hinge arms. A push-pull rod from each aileron, connects to the bellcrank lever in the corresponding fore flap hinge arm. Refer to Figure 1 for details.

Because of the method of attachment, the ailerons droop with the wing flaps, and the geometric arrangement of the control linkage is such that the aileron range of movement increases proportionally as the flaps go down. The ailerons move differentially at any flap position.

Each aileron carries a geared tab, and the left aileron carries an electrically-operated trim tab (refer to 27-13-00, Aileron Trim Tab System – General Data and 27-16-00, Aileron Geared Tab System – General Data).

Note For general cable rigging instructions refer to 27-00-00, Flight Controls – General Data Para 2.A.



Aileron System
Figure 1

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2. Adjustment/Test

A. Rig Aileron Control System

Refer to Figure 2 and Figure 3 for details.

Note Before attempting to rig the aileron control system ensure that wing flap system is correctly rigged (refer to 27-50-00, Wing Flaps System – General Data).

1) Pump flaps to fully up position.

2) Disconnect spring strut from co-pilot's handwheel.

3) Install rigging pin SD10544 (0.25 inch dia.) in base of control column.

4) Check that axis of each handwheel is horizontal. If necessary, remove covers from control column arms and adjust chain turnbuckles until handwheels are aligned. Set chain tension to value shown in Figure 2.

5) Check aileron quadrant for neutral setting by installing rigging pin SD10543 (0.375 inch dia.). Adjust cable circuit turnbuckles, if necessary, to obtain neutral position, maintaining correct cable tension (refer to 27-00-00, Flight Controls – General Data).

6) Check neutral position of aileron linkage by inserting rigging pin SD10542 (0.25 inch dia.) in linkage pulleys. Adjust circuit turnbuckles, if necessary, to obtain neutral position, maintaining correct cable tension (refer to 27-00-00, Flight Controls – General Data).

7) Remove all rigging pins and operate system through several cycles, then check that all rigging pins enter rigging holes freely.

8) With rigging pin installed in aileron linkage pulleys, check that trailing edge of aileron aligns with trailing edge of flap; adjust push-pull rod at aileron if necessary.

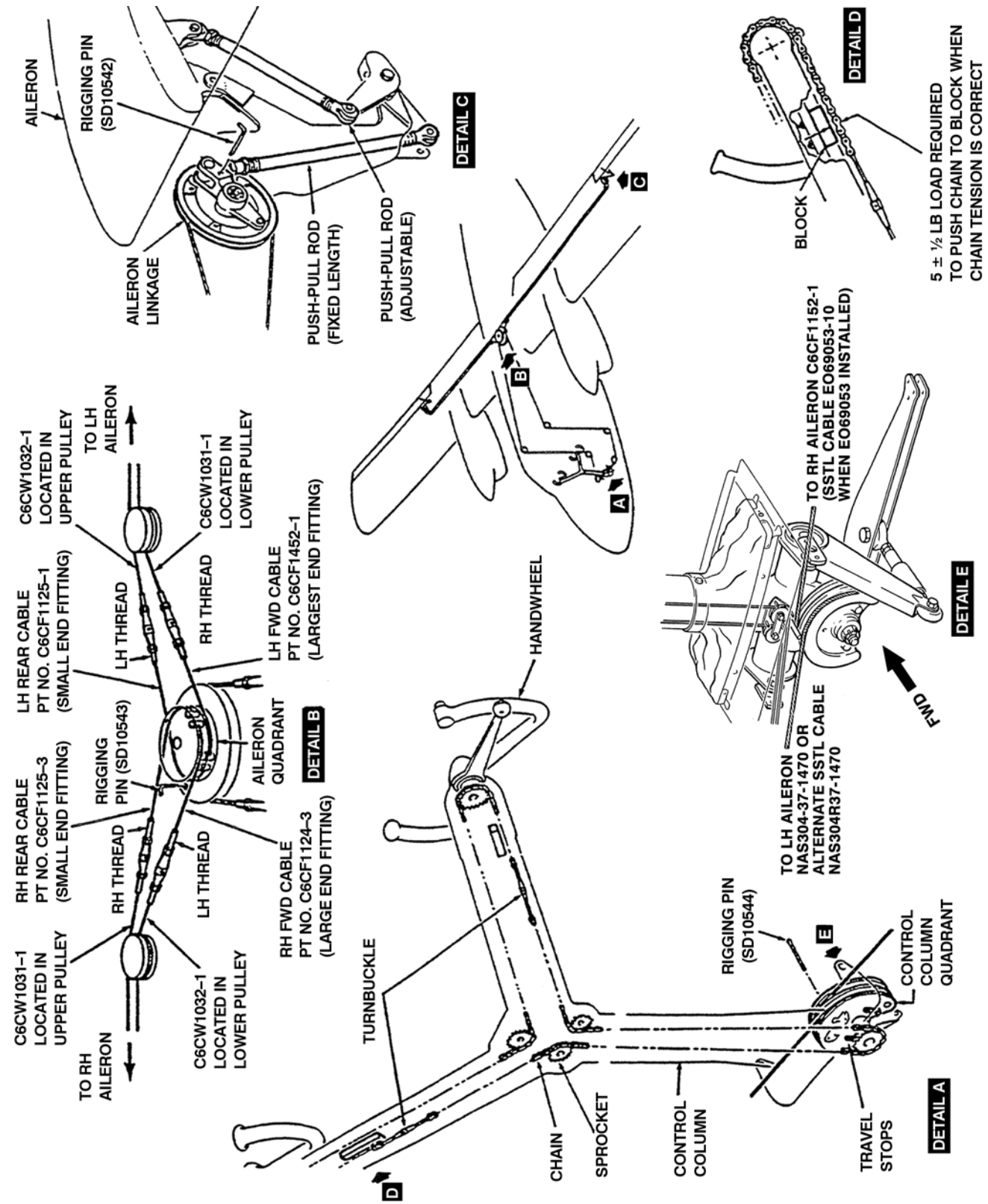
9) Remove rigging pins and operate controls over full range of travel with flaps in up position, then in down position. Adjust stops at base of control column, if necessary, to obtain desired travel (refer to Figure 3).

Note Flaps must be in fully up position when checking aileron travel.

10) Reconnect spring strut to co-pilot's handwheel. Adjust, if necessary, to maintain handwheel neutral position with no spring load.

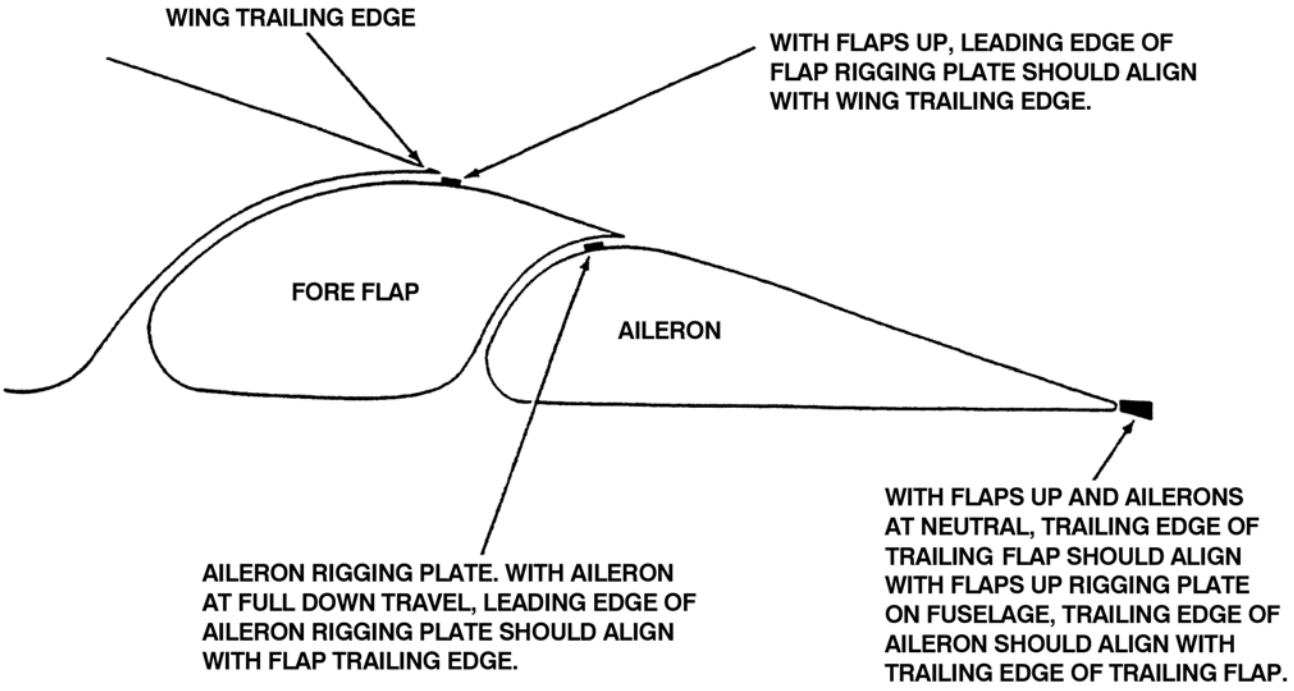
11) Check aileron control surface clearances (refer to 27-00-00, Flight Controls – General Data), and aileron geared tab rigging (refer to 27-16-00, Aileron Geared Tab System – General Data).

12) Operate ailerons and check for full and free range of movement.



Aileron System – Rigging
Figure 2

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Aileron Rigging Marks
Figure 3

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